

API PIPES ERW & SEAMLESS PIPES

API MECHANICAL & CHEMICAL SPECIFICATION

TENSILE AND CHEMICAL REQUIREMENTS (EXCERPTS FROM API SPECIFICATIONS, 1983 EDITION)

Application	Grade	Tensile Requirements						Elongation	
		Yield Strength			Tensile Strength				% min
		psi min	MPa min	kg/mm ² min	psi min	MPa min	kg/mm ² min		
Line Pipe	API 5L Grade A	30,000	207	21.1	48,000	331	33.7	e = 625,000 x $\frac{A^{0.2}_{*2}}{U^{0.9}}$	
	API 5L Grade B	35,000	241	24.6	60,000	413	42.2		
	API 5L Grade X42	42,000	289	29.5	60,000	413	42.2		
	API 5L Grade X46	46,000	317	32.3	63,000	434	44.3		
	API 5L Grade X52	52,000	358	36.6	(1) 66,000* ¹ (2) 72,000	(1) 455 (2) 496	46.4 50.6		
	API 5L Grade X56* ⁶ 56,000	386	39.2	(1)	71,000* ¹ (2) 75,000	(1) 489 (2) 517	49.9 52.7		
	API 5L Grade X60* ³ 60,000 * ⁶	413	42.2	(1)	75,000* ¹ (2) 78,000	(1) 517 (2) 537	52.7 54.8		
	* ⁶ API 5L Grade X65* ⁸ 65,000 * ¹¹	448	45.7	(2)	(1) 77,000* ¹ 80,000	(1) 530 (2) 551	54.1 56.2		
	API 5L Grade X70* ⁶ 70,000 * ¹¹	482	49.2		82,000	565	57.6		
	API 5LU Grade X8080,000	551	56.2		95,000/ * ⁴ 125,000	655/ 862	66.8/ 87.8		
API 5LU Grade X100100,000	689	70.3		110,000/ * ⁴ 135,000	758/ 931	77.3/ 94.9			

- *1 (1) Wall thickness > 0.375 in., OD ≥ 20 in.
Any wall thickness, OD < 20 in.
(2) Wall thickness ≤ 0.375 in., OD ≥ 20 in.

- *2 The minimum elongation in 2 inches (50.8 mm) shall be determined by the following formula:

$$e = 625,000 \times \frac{A_{*2}^{0.2}}{U^{0.9}}$$

where:

e = minimum elongation in 2 inches (50.80 mm) in per cent rounded to the nearest 1/2 per cent.

A = cross sectional area of the tensile test specimen in square inches, based on specified outside diameter or nominal specimen width and specified wall thickness rounded to the nearest 0.01 sq. in., or 0.75 sq. in., whichever is smaller.

U = specified tensile strength, psi.

- *3 The minimum tensile strength for grade X60 electric-resistance welded pipe in all sizes and wall thicknesses shall be 75,000 psi (52.7 kg/mm²).

- *4 Tensile strength means min. value/max. value.

- *5 Figures in parentheses are values for seamless pipe. Upper line value means non-expanded. Lowerline value means cold-expanded. In grades X42 through X65, for each reduction of 0.01 per cent below the maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible, up to a maximum of 1.45 per cent for X52 and lower and up to a maximum of 1.60 per cent for grades higher than X52.

- *6 Other chemical analyses may be furnished by agreement between purchaser and manufacturer.

- *7 Either columbium, vanadium, titanium, or a combination thereof, shall be used at the discretion of the manufacturer.